



SS304 Micro-interlayered Low Alloy Steel Prepared via Cold Metal Transfer Based Wire Arc Additive Manufacturing

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Abstract

This study investigates the effect of incorporating an SS 304 micro-interlayer (SS304-IL) in low alloy steel (ER70S6) fabricated using Cold Metal Transfer (CMT)-based Wire Arc Additive Manufacturing (WAAM) to enhance mechanical properties, microstructure, and corrosion resistance. Multi-layered samples were produced by manually introducing a 30 µm-thick SS 304 thin film between successive ER70S6 layers, allowing the molten deposit to fuse the interlayer with the base material. Microstructural analysis revealed improved grain structure uniformity, with a slight increase in grain size from 8.52 µm (ER70S6) to 9.21 µm (SS304-IL) and an increase in ferrite content from 80.11 to 81.74%. The density of the interlayered sample increased to 7700 kg/m³, while porosity was reduced from 1.84% to 1.55%, enhancing material integrity. Tensile testing in different orientations showed a reduction in anisotropy from 12.41% (ER70S6) to 3.54% (SS304-IL), indicating improved structural reliability. Hardness exhibited a slight increase from 177 HV to 180 HV. Corrosion resistance was assessed using electrochemical impedance spectroscopy (EIS), revealing a higher corrosion potential (E_{corr}) of -0.662 V and a lower corrosion current density (I_{corr}) of -6.3006 A/cm² in SS304-IL, compared to -0.66562 V and -6.1283 A/cm² in ER70S6, respectively. These findings demonstrate that the SS 304 interlayer improves mechanical properties, reduces anisotropy, and enhances corrosion resistance, making it a promising approach for optimizing WAAM-fabricated components.

Keywords Wire arc additive manufacturing (WAAM) · Cold metal transfer (CMT) · Low alloy steel (ER70S6) · SS 304 interlayer · Corrosion

Why WAAM?

Introduction

The manufacturing landscape has undergone a profound transformation, shifting from traditional subtractive techniques to the innovative domain of Additive Manufacturing (AM), commonly known as 3D printing. Unlike conventional manufacturing, which involves material removal, AM builds structures layer by layer from digital models, enabling the fabrication of intricate and highly precise components [1]. This technology has gained widespread attention due to its ability to produce complex geometries without the need for additional tooling, such as casting moulds or dies, making it highly efficient and customizable [2, 3]. Additionally,

AM reduces material waste and production time, making it a cost-effective solution for industries requiring high-performance and lightweight structures [2, 4]. These advantages have led to its adoption across various fields, including aerospace, defence, healthcare, electronics, automotive, civil engineering, and ocean engineering [5]. In aerospace, AM enables the production of lightweight components that improve fuel efficiency, while in healthcare, it facilitates the development of patient-specific implants and prosthetics. Similarly, the automotive industry employs AM for rapid prototyping and customized low-volume parts [3, 6].

Among the various metal AM techniques, wire arc additive manufacturing (WAAM) has emerged as a promising method due to its ability to produce large, structurally robust metal parts at a high deposition rate. As a directed energy deposition (DED) method, WAAM utilizes an electric arc as a heat source to melt and deposit metal wire layer by layer [7]. This process is particularly advantageous for applications in industries such as aerospace, automotive, and maritime, where large-scale metal components with high

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mechanical strength are required [8]. WAAM is commonly implemented using metal inert gas (MIG), tungsten inert gas (TIG), and Plasma Arc Welding (PAW), with MIG being the most widely used due to its efficiency and continuous wire feed mechanism [9, 10]. A notable advancement within WAAM is Cold Metal Transfer (CMT), a modified MIG process that offers enhanced control over heat input, resulting in reduced spatter, improved grain structure, and superior mechanical properties [11]. By synchronizing wire motion with electrical process control, CMT enables the fabrication of high-quality WAAM components with finer microstructures and reduced thermal distortion [9, 12, 13]. However, despite its advantages, WAAM-fabricated components often exhibit microstructural inhomogeneity, anisotropy, and porosity, which can compromise mechanical performance and structural reliability.

To address these challenges, various post-processing and in-process modifications have been explored. Heat treatments have been employed to relieve residual stresses and enhance material strength, while cooling techniques such as near immersion active cooling (NIAC) and interpass CO₂ cooling have been developed to control thermal accumulation and refine grain structures [14–16]. Interpass cold rolling and Hot Forging WAAM (HFWAAM) have also been introduced to reduce anisotropy and residual stresses by inducing plastic deformation during the deposition process [17–19]. While these methods have demonstrated improvements in WAAM-fabricated components, they increase processing time and cost, making it imperative to explore alternative strategies that enhance mechanical properties and structural integrity during the deposition process rather than relying solely on post-processing techniques.

One promising yet unexplored approach is the incorporation of micro/nano interlayers between deposited layers in WAAM. The introduction of a micro-interlayer presents a novel strategy for improving microstructural homogeneity, mechanical properties, and corrosion resistance without requiring extensive post-processing. This study investigates the effect of incorporating a thin SS 304 micro-interlayer between successive layers of low carbon steel (ER70S6) fabricated using CMT-based WAAM. The SS 304 interlayer is expected to refine the grain structure, reduce anisotropy, and enhance material integrity. A comprehensive evaluation of mechanical properties, microstructure, and corrosion

resistance is conducted, comparing SS 304 interlayer specimen (SS304-IL) with standard ER70S6 specimens. By integrating micro-interlayers, this research aims to introduce an efficient and cost-effective method for improving WAAM-fabricated components, addressing key challenges in metal additive manufacturing

Experimental

The WAAM process was carried out using a six-axis YASKAWA AR1440 welding robot in conjunction with a Fronius TPS 500i CMT welding machine. A rectangular specimen with dimensions of 160 mm in length and 80 mm in height was built layer by layer, achieving a thickness of approximately 8 mm. The feedstock used for this process was copper-coated ER70S6 wire, with a diameter of 1.2 mm. The interlayer material utilised in the manufacturing process was SS 304, with a thickness of 30 microns. Thin foils were cut into rectangular pieces measuring 180 mm in length and 20 mm in width. The deposition was performed on a mild steel substrate with a thickness of 10 mm. The composition of the feedstock and SS 304 is shown in Table 1.

In this study, two sets of CMT-WAAM samples were prepared using low carbon steel (ER70S6). One set of standard ER70S6 specimen without any interlayer and another set incorporating a 30 µm-thick SS 304 thin film, SS304-IL specimen. The 30 µm thickness was chosen for interlayer as it was the thinnest available size, ensuring ease of handling and manipulation during the manufacturing process. Additionally, a thinner interlayer was expected to fuse and integrate more effectively with the base material, promoting better metallurgical bonding. The ER70S6 specimen was prepared by depositing multiple layers of the base material one above the other onto a mild steel substrate. The SS304-IL specimen was produced by placing a thin sheet of SS 304 above each deposited layer before adding the next layer. As the new layer is added, the interlayer material fuses with the base material. This sequence of adding interlayers and depositing new layers continued until the specimen reached the desired dimensions. The interpass temperature was maintained below 350 °C, as higher temperatures led to the collapse of the deposited material. A schematic of the deposition strategy is shown in Fig. 1a–c. The first layer, with the

Table 1 Composition of ER70S6 and SS 304

Sample	Elements								
	C (%)	Mn (%)	Si (%)	S (%)	P (%)	Cu (%)	Cr (%)	Ni (%)	Fe (%)
ER70S6	0.06–0.14	1.40–1.60	0.80–1.0	0.025 (max)	0.025 (max)	0.50 (max)	0.0	0.0	Balance
SS 304	0.08	≤ 2.00	≤ 1.00	≤ 0.030	≤ 0.045	0.0	18–20	8–11	Balance

interlayer positioned between, was deposited from left to right as shown in Fig. 1b. The next layer was then deposited from right to left. This alternating deposition process was repeated until the desired specimen dimensions are obtained. The optimal welding parameters were determined through trial and error to achieve a uniform weld profile. Visual inspection was carried out to assess the specimen's integrity, ensuring it was free from defects such as cracks, porosity, and insufficient fusion. The finalized welding parameters were 120 A current, 18.5 V voltage, and a wire feed speed of 4 m/min. The shielding gas was a mixture of 80 % Argon and 20 % CO₂, with a flow rate set to approximately 15 LPM.

The prepared WAAM samples were finished using CNC milling process. Specimens with the required dimensions and geometry were prepared for various testing/characterization, using Wire Electrical Discharge Machining (WEDM), as demonstrated in Fig. 2. For microstructural characterization, specimens (10 × 10 mm) were taken from the top (T) and bottom (B) of the WAAM samples. Tensile test samples were prepared according to the ASTM E8M standard and were cut in vertical (V), diagonal (D), and horizontal (H) orientations. For corrosion testing, square samples (C) with 20 mm sides and 5 mm thickness was cut from the central area of the specimen

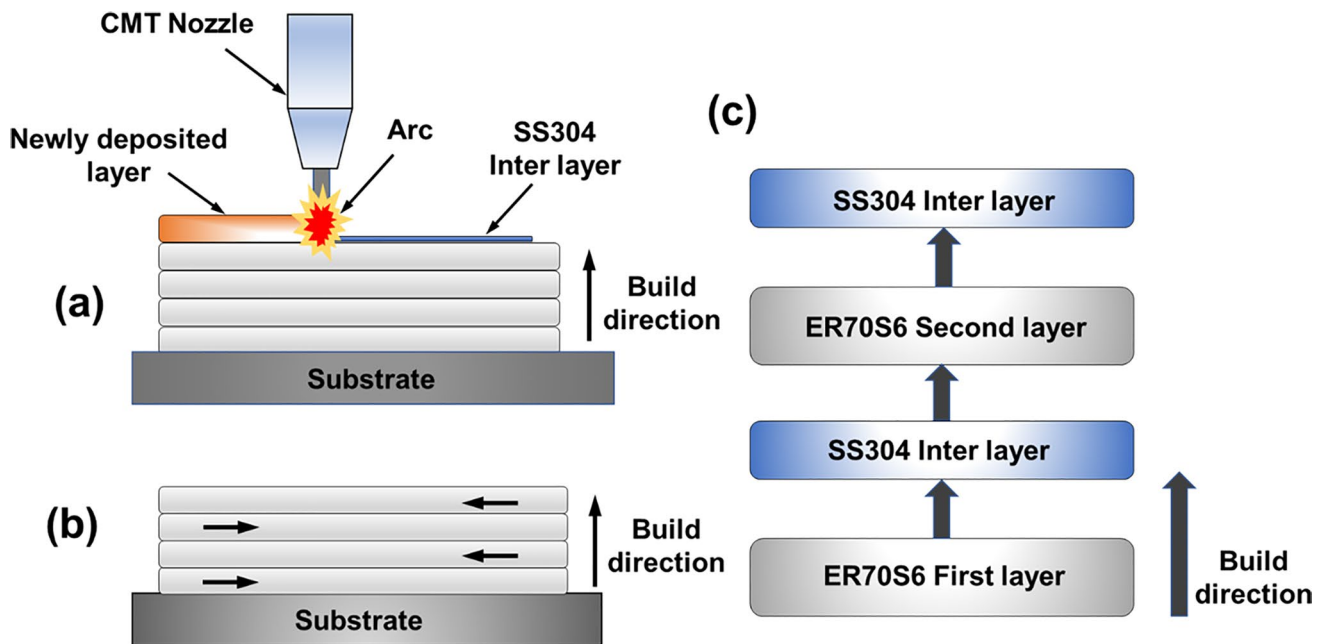


Fig. 1 a Schematic diagram of WAAM process with interlayer. b Deposition strategy. c Flowchart of WAAM process with interlayer

Fig. 2 Sample preparation for tensile, microstructure, and corrosion tests

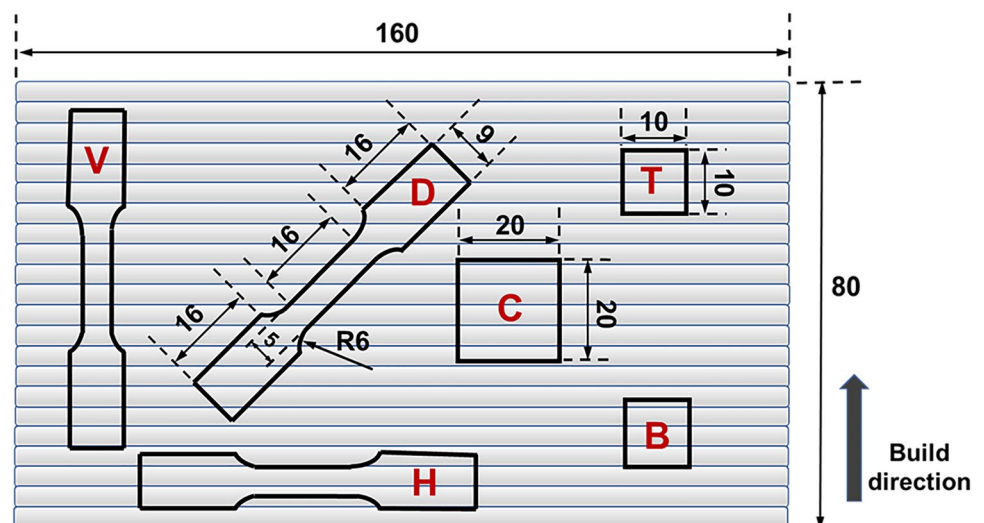
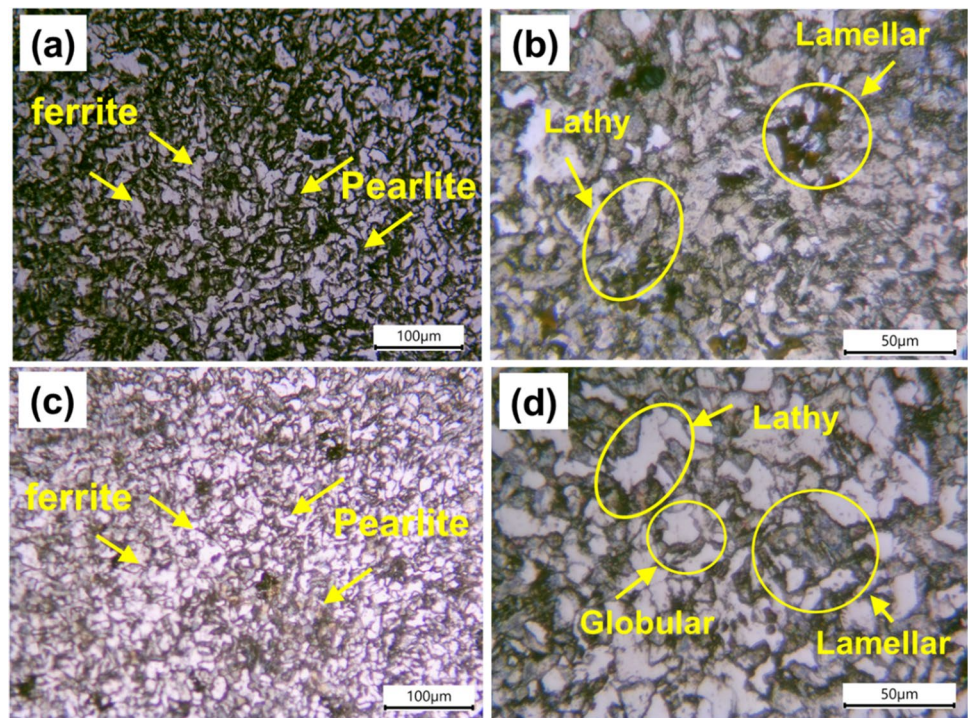


Fig. 3 Optical micrographs of **a, b** ER70S6 (top) at 100 μm and 50 μm magnifications, and **c, d** SS304-IL (top) at 100 μm and 50 μm magnifications



The density of the WAAM samples was measured using density measurement kit based on Archimedes' principle. The weight of each sample was recorded both in air and submerged in water, and the density was then calculated using the corresponding formula.

$$\frac{\text{Density(Sample)}}{\text{Density(Water)}} = \frac{\text{Wt. of Sample(Air)}}{\text{Wt. of sample(Air)} - \text{Wt. of sample(Water)}}$$

For optical microscopy, the samples were polished to a mirror-like finish and etched using Nital solutions containing up to 5 vol. % concentrated nitric acid. The etched specimens were examined at various magnifications using an Olympus BX53M optical microscope (Olympus Corporation, Japan). The elemental composition of the materials was determined using spark optical emission spectrometry, SPECTROMAXx (SPECTRO Analytical Instruments GmbH, Germany).

For SEM analysis, the samples were polished using various grades of silicon carbide paper to achieve a mirror-like surface. A ZEISS GeminiSEM 360 field emission scanning electron microscope (FESEM) equipped with ZEISS Smart EDX (Carl Zeiss AG, Germany) was used. X-ray diffraction (XRD) analysis was conducted using an X'Pert3 Powder X-ray diffractometer (Malvern Panalytical, Netherlands), operating at 45 kV and 40 mA with Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$).

Tensile tests were performed using a Shimadzu AG-X Plus universal testing machine (Shimadzu Corporation, Japan) with a load capacity of 10 kN. The vertical (V)

sample followed the build direction, the horizontal (H) sample was cut perpendicular to the build direction, and the diagonal (D) sample was cut at a 45-degree angle to the build direction. Flat dog bone specimens measuring 1.5 mm in thickness, 16 mm in gauge length, and 5 mm in cross-section, as shown in Fig. 2, were used. Tensile testing was conducted at a displacement control rate of 1 mm/min. From the tensile test data, stress-strain curves were generated, and parameters such as elongation to failure and ultimate tensile strength (UTS) were calculated.

Hardness was evaluated using the Vickers microhardness test. A Shimadzu HMV-G20S microhardness tester (Shimadzu Corporation, Japan) was used, with a load of 1 kgf and a dwell time of 10 seconds. A linear hardness profile was plotted along the build direction at 2.5 mm intervals, with measurements taken from the bottom to the top of the sample, and the average microhardness value was calculated.

The corrosion behaviour of the WAAM samples was assessed using electrochemical impedance spectroscopy (EIS). The experiments were conducted in a 3.5 wt.% NaCl solution at ambient temperature. The electrochemical setup included a calomel reference electrode immersed in a 0.1 mol/L KCl solution and a platinum counter electrode in a three-electrode cell configuration. The corroded surfaces were analysed using Alicona InfiniteFocus G5 (Bruker Alicona, Austria).

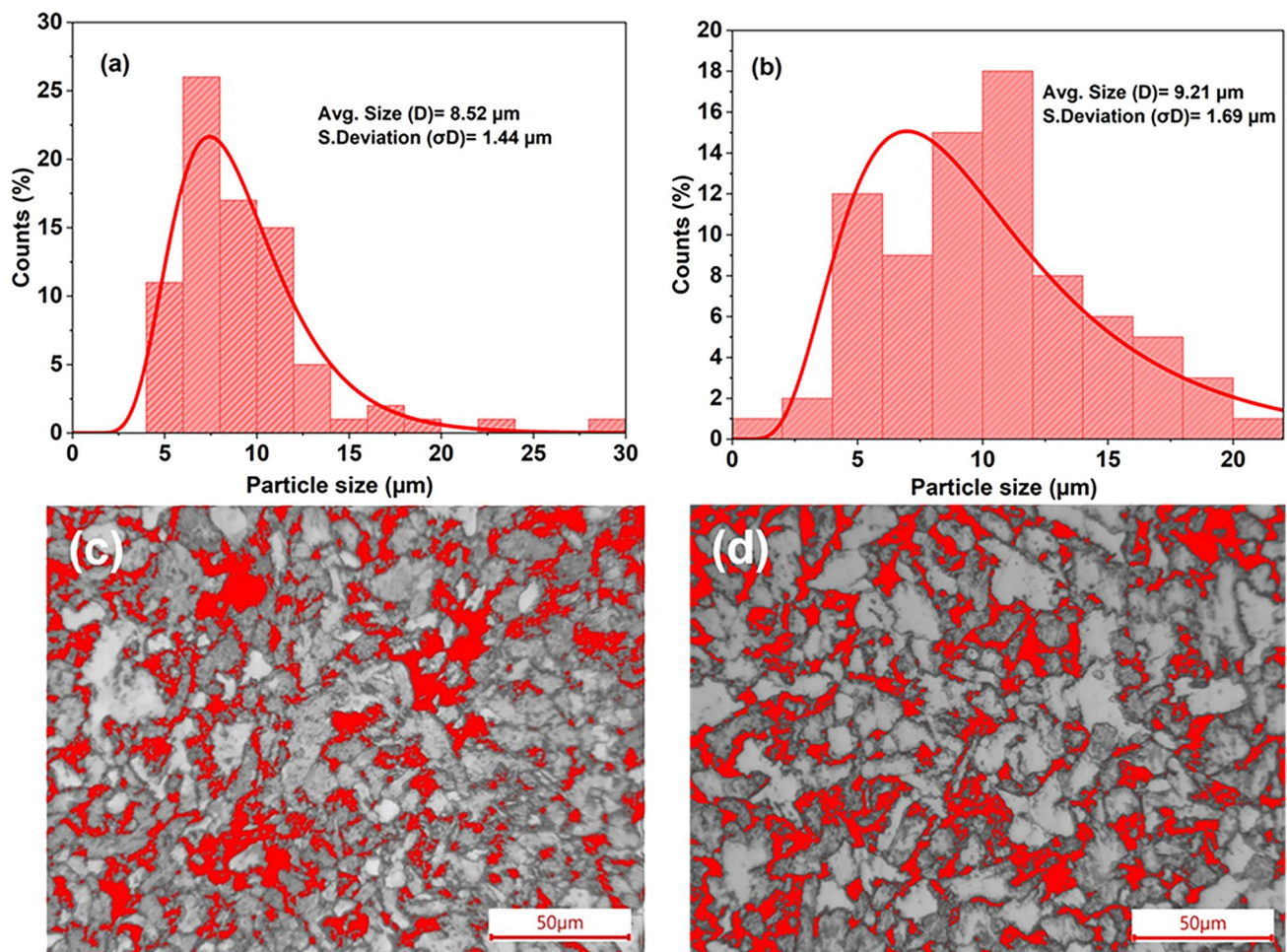


Fig. 4 **a, b** Grain size distribution of ER70S6 and SS304-IL samples fitted with a log-normal distribution (top). **c, d** Phase fraction analysis micrographs of ER70S6 and SS304-IL samples (top)

Results and Discussions

Microstructural Analysis

The macroscopic morphology of the WAAM-CMT specimens at 100 μm and 50 μm scales is shown in Fig. 3a–d. Various morphologies of δ ferrite and pearlite are visible in the structure. The microstructure of the steel consists of fine polygonal ferrite and a small volume fraction of lamellar pearlite, along with additional constituents such as bainite and acicular ferrite [20, 21]. The optical micrographs illustrate the distribution of the pearlite phase along the ferrite grain boundaries and show areas of non-equilibrium phases near fusion boundaries, which are attributed to varying cooling rates during solidification [22–24]. It is well known that in metal additive manufacturing processes, the presence of multiple and complex thermal cycles is the main factor leading to non-uniform microstructure along the building direction. A 10-minute interval between each layer can reduce

the effects of heat accumulation and prevent microstructural variations across different layers in the z-direction. Consequently, all layers experience similar heating and cooling cycles, resulting in a uniform microstructure along the building direction [18, 19].

The grain size distribution of the samples was estimated by fitting a grain size histogram with a log-normal function. To ensure accuracy, the dimensions of 80 individual grains were carefully measured from optical micrographs using ImageJ software. These measurements were then used to construct a histogram that visually depicts the grain size distribution in the sample, with the histogram data fitted to a log-normal function, $f(D) = \frac{1}{\sqrt{2\pi}\sigma D} \exp\left[-\frac{(D/D_0)^2}{2\sigma^2}\right]$, where D represents the average grain size, while σD denotes the standard deviation of the average grain size [25, 26]. The typical fitting of the log-normal distribution function to the grain size distribution histogram is shown in Fig. 4a and b for the standard ER70S6 specimen and the SS304-IL

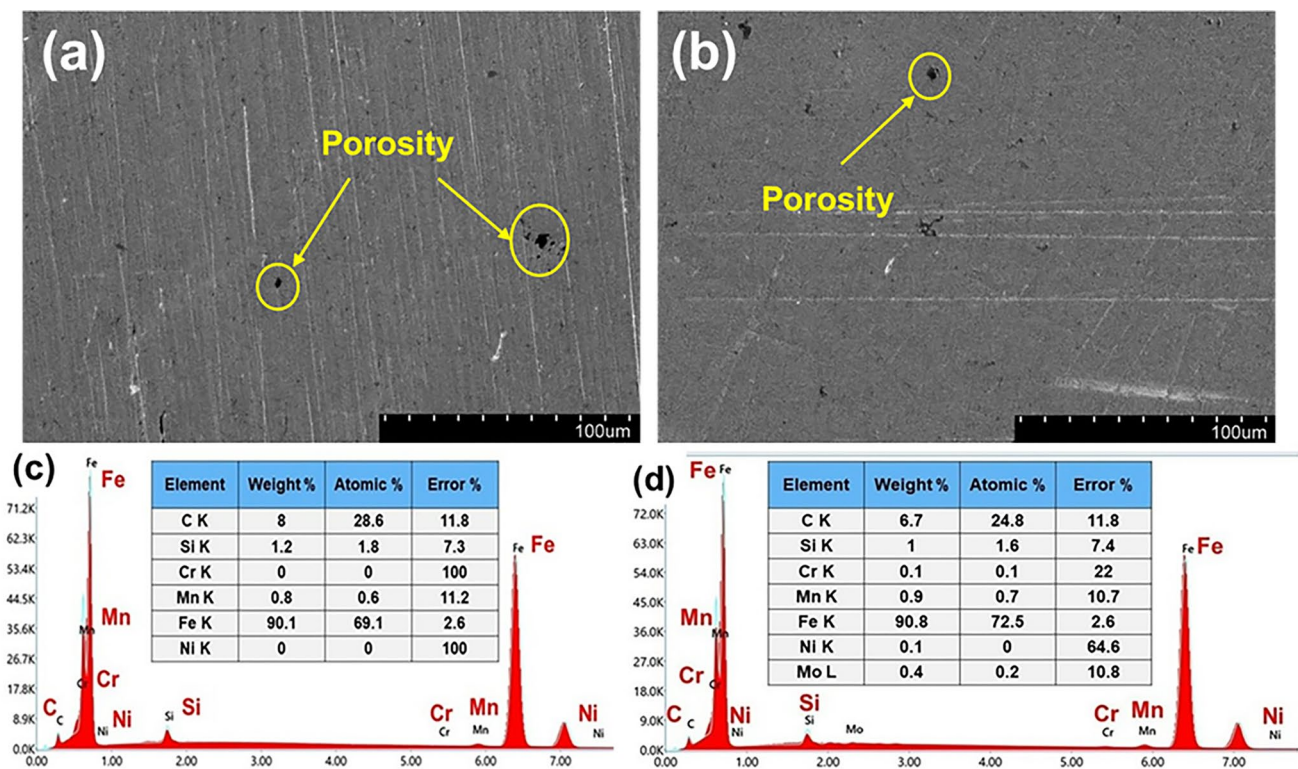


Fig. 5 **a** SEM image of ER70S6 (top), **b** SEM image of SS304-IL (top), **c** EDS data of WAAM ER70S6 (top), **d** EDS data of WAAM SS304-IL (top)

specimen (SS304-IL), respectively. The average grain size for the standard ER70S6 specimen was calculated to be 8.52 μm , whereas the SS304-IL specimen had an average grain size of 9.21 μm , indicating that the addition of the SS304-IL results in an increase in grain size on average. This could be due to heat accumulation caused by the lower thermal conductivity of SS 304 interlayer compared to ER70S6, leading to slower heat dissipation, reduced cooling rates, and prolonged exposure to elevated temperatures, which promote grain growth. It can be noted that the distribution of grain size is more non-uniform in the SS304-IL specimen. This could be due to the thermal barrier effect of the SS 304 interlayer, causing uneven heat dissipation and localized cooling rate variations. Additionally, elemental diffusion from SS 304 into ER70S6 may alter nucleation and grain growth dynamics, further contributing to the non-uniformity.

Figure 4c–d shows the phase fraction analysis micrograph, which illustrates the predominant microstructure at the core of each molten pool. This structure is characterized mainly by small polygonal ferrite (F) as the principal component, along with a minor proportion of the lamellar pearlite (P) phase, which is primarily located along the boundaries of the ferrite grains [27–29]. The percentage fractions of δ ferrite and pearlite were calculated using ImageJ software.

The amount of pearlite in the ER70S6 specimen is 19.89%, while it is 18.26% in the SS304-IL specimen [22, 30].

SEM and EDS Analysis

Figure 5a, b shows the SEM images of the polished surfaces of the WAAM specimens, both without an interlayer and with SS 304 interlayer. The images reveal that the specimens are free from defects such as lack of fusion and cracks; however, a small amount of porosity was observed in both samples. The porosity percentage, calculated from the SEM images, was found to be 1.55% for the SS304-IL specimen, compared to 1.84 % in the standard specimen. This decrease in porosity and subsequent increase in density of SS304-IL sample is in corroboration with the actual density measurements using Archimedes principle. The density of WAAM specimen, ER70S6 is 7672 kg/m^3 and that of SS304-IL specimen is 7700 kg/m^3 .

EDS analysis was performed on both specimens to identify the elements present, and elemental mapping was carried out to observe the distribution of the major elements. The chemical compositions of both samples are shown in Fig. 5c, d. The results indicated that the composition of the WAAM-printed ER70S6 closely matches that of the standard material. In the SS304-IL specimen, additional elements

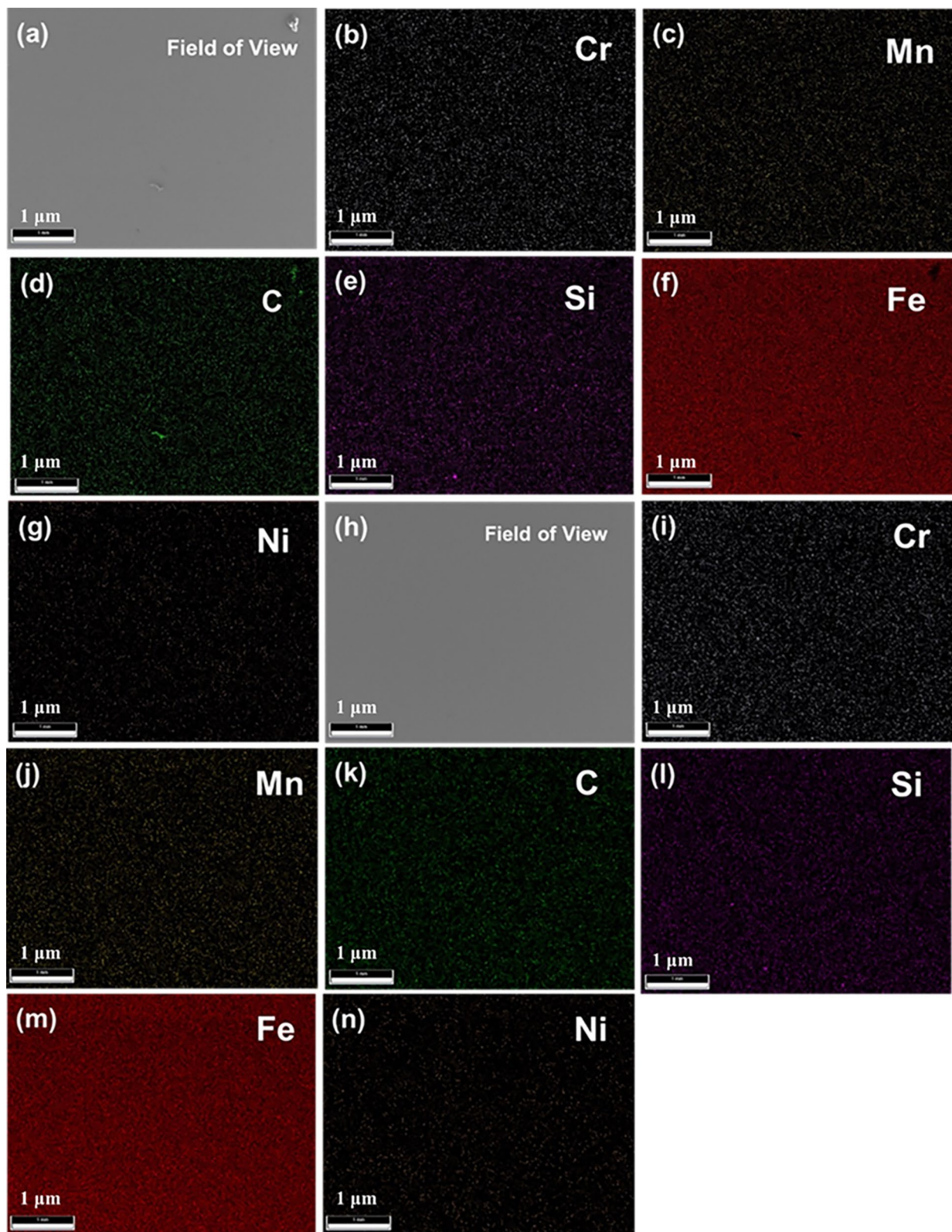
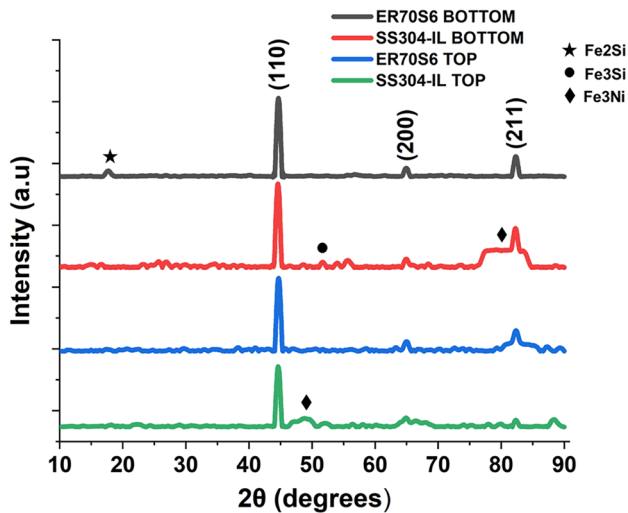


Fig. 6 a–g Elemental mapping of ER70S6 (top), h–n Elemental mapping of SS304-IL (top)

Table 2 Elemental compositional data of SS304-IL and ER70S6 obtained from OES

Wt. %	C	Mn	Si	S	P	Cu	Cr	Ni
SS304-IL	0.137	1.26	0.673	0.010	0.022	0.064	0.15	0.065
ER70S6	0.141	1.45	0.812	0.018	0.024	0.083	0.02	0.03

**Fig 7.** XRD pattern of ER70S6 specimen (Bottom), SS304-IL Specimen (Bottom), ER70S6 Specimen (Top), SS304-IL Specimen (Top)

such as nickel and chromium were present, suggesting that the interlayer successfully fused with the base material [25, 31].

Elemental mapping of the major elements was performed to determine their distribution patterns in both specimens, with the results shown in Fig. 6a–n. Both specimens, primarily composed of Fe, contain elements such as Si, C, and Mn. However, the SS304-IL specimen uniquely exhibits trace amounts of Cr and Ni. The distribution of Cr and Ni across the sample surface is relatively uniform, indicating a homogeneous dispersion within the material. These elements are evenly spread without significant clustering or localized concentration variations. The presence of Cr-rich regions may enhance corrosion resistance, while Ni-rich regions could potentially improve mechanical properties. The distribution of these elements may be correlated with the formation of specific microstructural features, such as carbides or nitrides.

Further, Spark optical emission spectroscopy (OES) was used to confirm the elemental composition of the specimens. The results for the SS304-IL specimen are shown in Table 2.

X-ray Diffraction (XRD) Analysis

Figure 7. presents the X-ray diffraction (XRD) patterns of the synthesized samples from both the top and bottom portions of the specimens along the build direction. The

diffraction peaks observed confirm the crystalline nature of the samples. Analysis reveals that the specimen is primarily composed of the α -Iron (Body-Centered Cubic, BCC, Ferrite) phase, with the main α -Iron peaks appearing at 2θ values of approximately 44.67° , 65.02° , and 82.34° . The austenite (Face-Centered Cubic, FCC, Iron) phase was not detected, suggesting that retained austenite either did not form in the as-printed sample or is present in a volume fraction too low to be detected. Similarly, the cementite phase is absent in the XRD spectra due to its significantly lower volume fraction compared to the ferrite phase [22, 32].

The diffraction peaks were indexed to the cubic phase based on ICDD Reference Code 01-087-0721. Additional secondary phases, including Fe_2Si (★), Fe_3Si (●), and Fe_3Ni (◆), were identified, particularly in SS304-IL specimens.

The presence of Fe_3Ni intermetallics indicates diffusion of Ni from the SS 304 interlayer, contributing to microstructural evolution, which may influence mechanical properties such as hardness and ductility. Crystallite sizes were determined utilizing the Scherrer equation, yielding average dimensions of 9.76 nm for ER70S6 (Top), 11.49 nm for SS304-IL (Top), 14.23 nm for ER70S6 (Bottom), and 9.27 nm for SS304-IL (Bottom). Dislocation density calculations indicate that SS304-IL (Bottom) exhibits a significantly higher dislocation density (102.61 nm^{-2}) compared to the other samples: ER70S6 (Top) at 30.34 nm^{-2} , SS304-IL (Top) at 48.37 nm^{-2} , and ER70S6 (Bottom) at 4.99 nm^{-2} , suggesting an elevated concentration of defects in SS304-IL (Bottom) [33]. The micro strain values for all samples are relatively low, indicating minimal lattice distortion. Specifically, the micro strain values are 13.70 for SS304-IL (Bottom), 8.96 for ER70S6 (Top), 11.14 for SS304-IL (Top), and 4.96 for ER70S6 (Bottom).

Tensile Test

Tensile properties were measured in each of the three directions (H, D, V) relative to the build direction [34]. The standard stress versus strain diagrams were plotted and are shown in Fig. 8a, b. The ultimate tensile strength (UTS) of the specimen in the horizontal direction is the highest, measuring 532 MPa. The UTS values for the vertical and diagonal specimens are similar, with the vertical specimen having a slightly higher UTS of 476 MPa compared to the diagonal specimen, which has a UTS of 466 MPa. A similar pattern was observed in SS304-IL specimen, the horizontal specimen exhibited the highest UTS of 509 MPa, followed by the

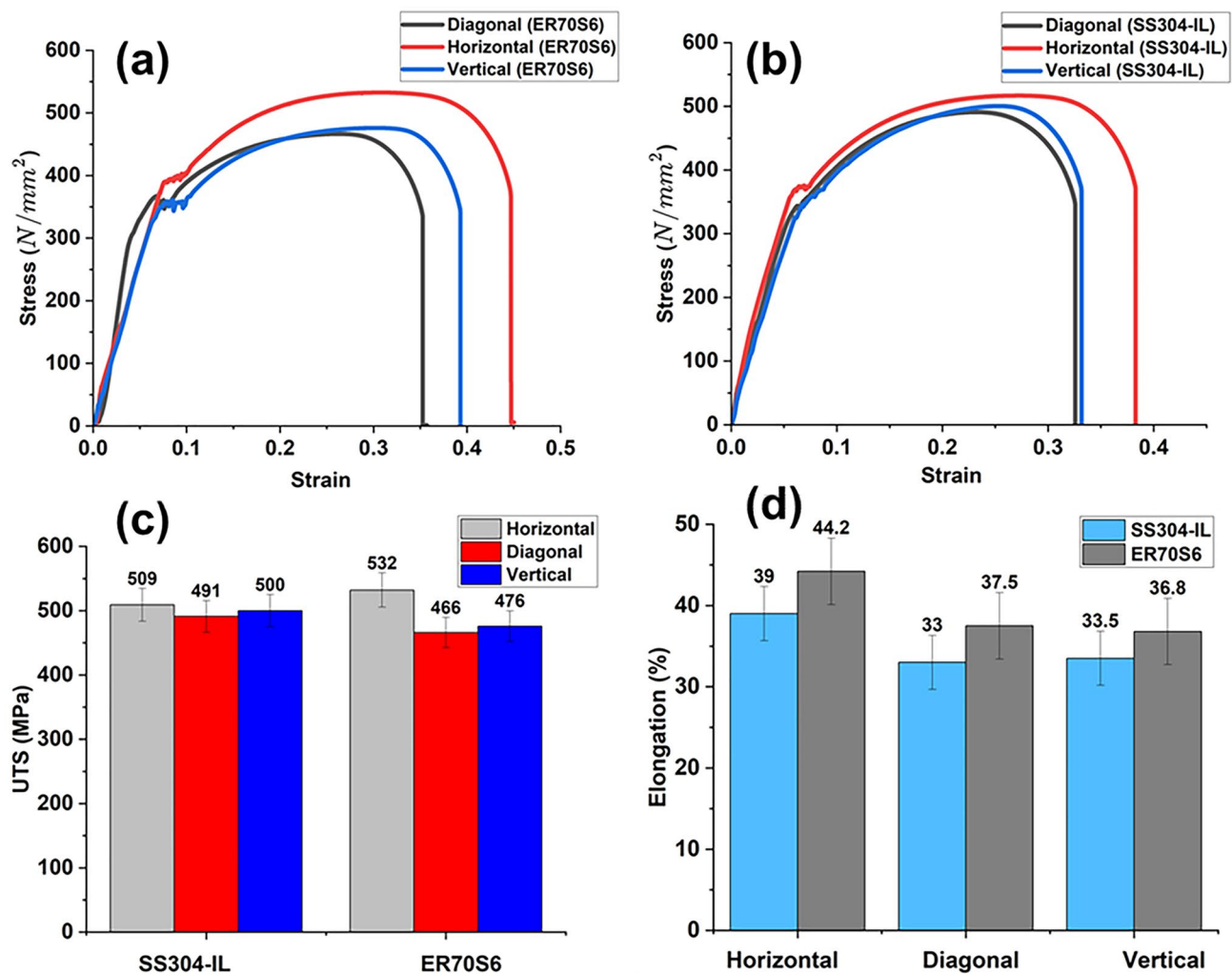


Fig. 8 **a** Representative stress-strain curve of ER70S6 in H, V, and D orientations. **b** Stress-strain curve of SS304-IL in H, V, and D orientations. **c** UTS comparison of ER70S6 and SS304-IL. **d** Elongation comparison of ER70S6 and SS304-IL

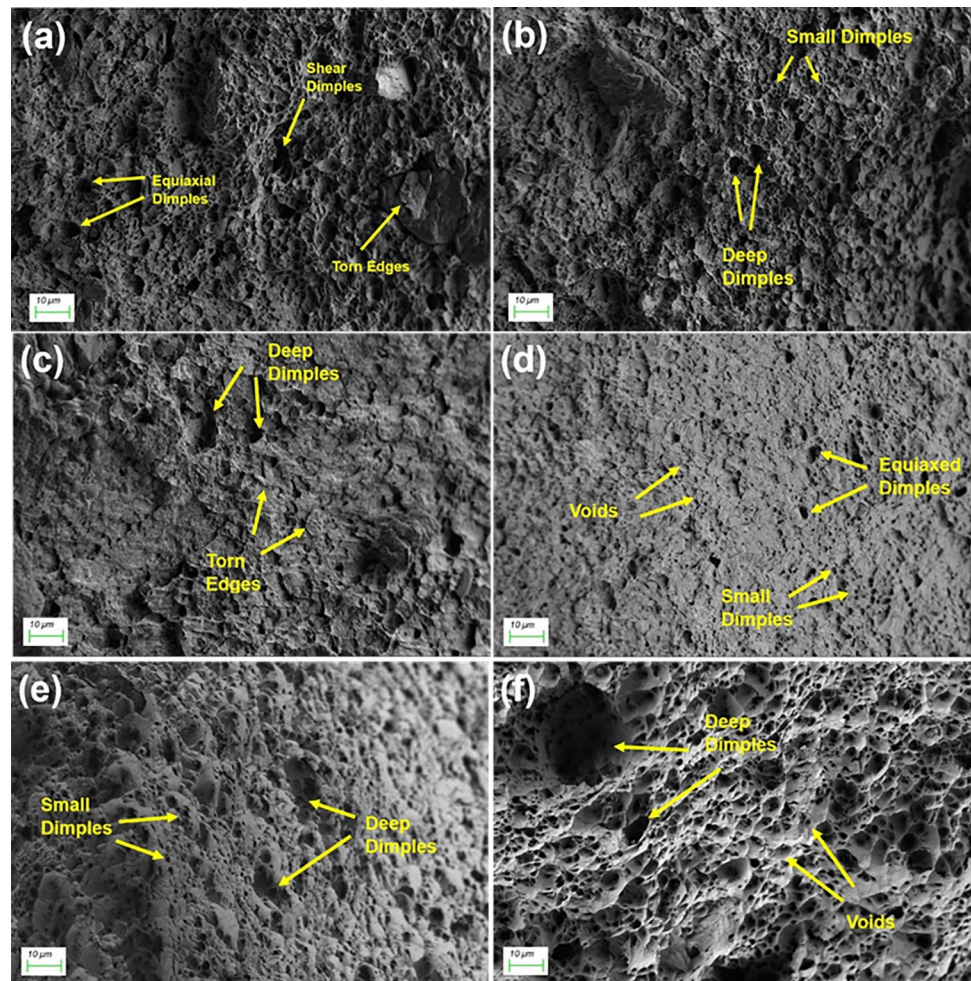
vertical specimen at 500 MPa, and the diagonal specimen at 491 MPa. On comparison, it can also be observed that the UTS of the ER70S6 specimen in the horizontal direction is higher than that of the SS304-IL specimen in the same orientation. However, for the diagonal and vertical specimens, the UTS values are higher in the SS304-IL specimen.

The calculated elongation values are shown in Fig. 8d. These values typically fall within the range expected for mild steel [35, 36] and the observed elongation values are on the higher side, indicating good ductility. Horizontally oriented specimens exhibited significantly greater elongation compared to those oriented vertically or diagonally. This discrepancy is more pronounced in the standard ER70S6 specimen than in the SS304-IL specimen. The elongation of the vertical and diagonal specimens is significantly lower than that of the horizontal specimens, highlighting a clear presence of anisotropy in ductility. Mechanical anisotropy in ductility

and toughness is commonly observed in additively manufactured parts and has been reported in studies using various metals and alloys [37]. Anisotropy can be calculated using the formula, $A = \left(\frac{P_{\max} - P_{\min}}{P_{\max}} \right) * 100$, where, A is the anisotropy and $P_{\max}$ and $P_{\min}$ are the maximum and minimum UTS values [33]. The anisotropy of ER70S6 specimen is 12.41 and that of SS304-IL specimen is 3.54. The values suggest that the anisotropy of have been reduced considerably while adding SS304 interlayer.

Figure 9 displays the SEM images of the fracture surfaces in the horizontal, diagonal, and vertical directions for both the standard ER70S6 specimen and the SS304-IL specimen. The specimens exhibit a significant number of equiaxed dimples in their microscopic morphology, indicating good plasticity. This plasticity is attributed to the unique directional grain growth that occurs during the arc deposition process.

Fig. 9 SEM fracture surface images of ER70S6 in different orientations: **a** horizontal, **b** diagonal, **c** vertical, and SS304-IL in different orientations: **d** horizontal, **e** diagonal, **f** vertical



However, the vertical SS304-IL specimen showed deeper dimples compared to the other specimens, the formation of deep dimples in vertical specimens is attributed to weak interlayer bonding and stress concentration at layer interfaces. During tensile loading, cracks initiate at these weak regions and propagate along interlayer boundaries, resulting in ductile fracture characterized by deeper dimples [38–40]

Microhardness

Vickers Microhardness testing was performed to compare the hardness values of the specimens. A linear hardness profile was measured from the bottom to the top of the sample in the build direction, with measurements taken at 2.5 mm intervals [39]. The hardness profile is plotted and shown in Fig. 10. The test was conducted with a load of 1 kgF and a dwell time of 10 seconds [41]. The results are displayed in the chart below. It was observed that the average hardness value of the SS304-IL specimen (180 HV) is slightly higher than that of the ER70S6 specimen (177 HV).

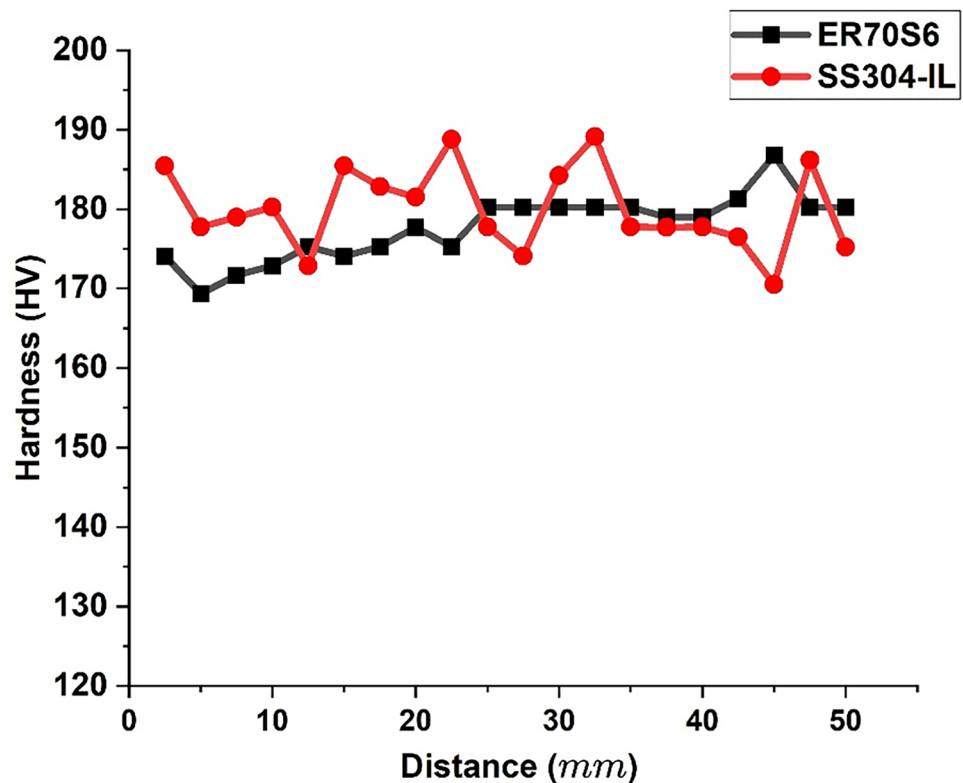
The hardness plot shows variations along the build direction for both ER70S6 specimen and SS304-IL specimens,

with the SS304-IL specimen exhibiting greater fluctuations. This increased variation is primarily due to the lower thermal conductivity of added SS 304 interlayer, which causes more heat accumulation, leading to uneven cooling rates, grain growth, and microstructural inconsistencies. In WAAM, thermal cycling and reheating of previously deposited layers result in grain coarsening in the heat-affected zone (HAZ), contributing to hardness fluctuations. ER70S6 specimen shows a relatively stable hardness trend, with minor fluctuations indicating a more uniform grain structure and consistent heat dissipation. In contrast, the SS304-IL specimen experiences significant variation due to inhomogeneous mixing of the interlayer material and localized thermal effects [22, 42].

Corrosion Test

Electrochemical testing was carried out to assess the corrosion behaviour of the prepared samples. The measurements were conducted at room temperature using a CHI608E Potentiostat/Galvanostat equipped with a standard three-electrode system. A 3.5 wt.% NaCl solution served as the

Fig. 10 Linear Hardness profile in built direction (Bottom to Top)



electrolyte. An Ag/AgCl electrode, immersed in a 3.5 wt.% KCl solution, functioned as the reference electrode, while a platinum electrode was used as the counter electrode, and the specimen acted as the working electrode [27, 43]. The working electrode sample was polished to a 2000 grit finish. Polarization curve analysis was used to determine the corrosion current density (I_{corr}) and corrosion potential (E_{corr}) values [44, 45]. The Tafel diagram (Fig. 11) illustrates the linear relationship between the logarithm of current and overall potential. The Tafel slope is linked to the activation energy of the process, with a steeper slope indicating higher activation energy and a slower reaction rate. The intersection point on the curve signifies the transition from anodic to cathodic behaviour, marking the equilibrium of oxidation and reduction reactions, which defines the corrosion potential. Beyond this point, oxidation or corrosion of the electrode material begins. The E_{corr} of the ER70S6 specimen is -0.66562 V, which is slightly lower than that of the SS304-IL Specimen (-0.662 V). The I_{corr} value of the ER70S6 specimen is -6.1283 A/cm², which is higher compared to the SS304-IL specimen (-6.3006 A/cm²). These results suggest that SS304-IL specimen has a lower corrosion rate than the ER70S6 specimen [46, 47].

The corroded surface was analysed using the Alicona InfiniteFocus G5. Figure 12a and b show the surface of

the ER70S6 specimen before and after corrosion testing, while Fig. 12c and d display the surface of the SS304-IL specimen before and after corrosion [48]. Observations indicate a reduction in pits, surface roughness, and corrosion product distribution in the SS304-IL specimen. The surface roughness (R_a) values were 1.058 μm and 1.8652 μm for the ER70S6 specimen before and after corrosion, respectively, and 1.069 μm and 1.4342 μm for the SS304-IL specimen. The higher R_a value of the corroded ER70S6 specimen compared to the corroded SS304-IL specimen indicates a relatively lower corrosion rate for the SS304-IL specimen.

The incorporation of an SS 304 interlayer enhances the corrosion resistance of the WAAM specimen by modifying its electrochemical behavior and preventing rapid material degradation. The presence of chromium in SS 304 promotes the formation of a stable passive oxide layer, which acts as a protective barrier against aggressive chloride ions in the electrolyte, reducing the overall corrosion rate. Additionally, nickel in the SS 304 interlayer stabilizes this passive film, minimizing localized corrosion such as pitting. The interlayer alters the anodic and cathodic reactions, leading to a lower corrosion current density (I_{corr}) and a slightly higher corrosion potential (E_{corr}) compared to the standard ER70S6 specimen. This improved electrochemical stability reduces the rate of metal dissolution, resulting in enhanced corrosion resistance.

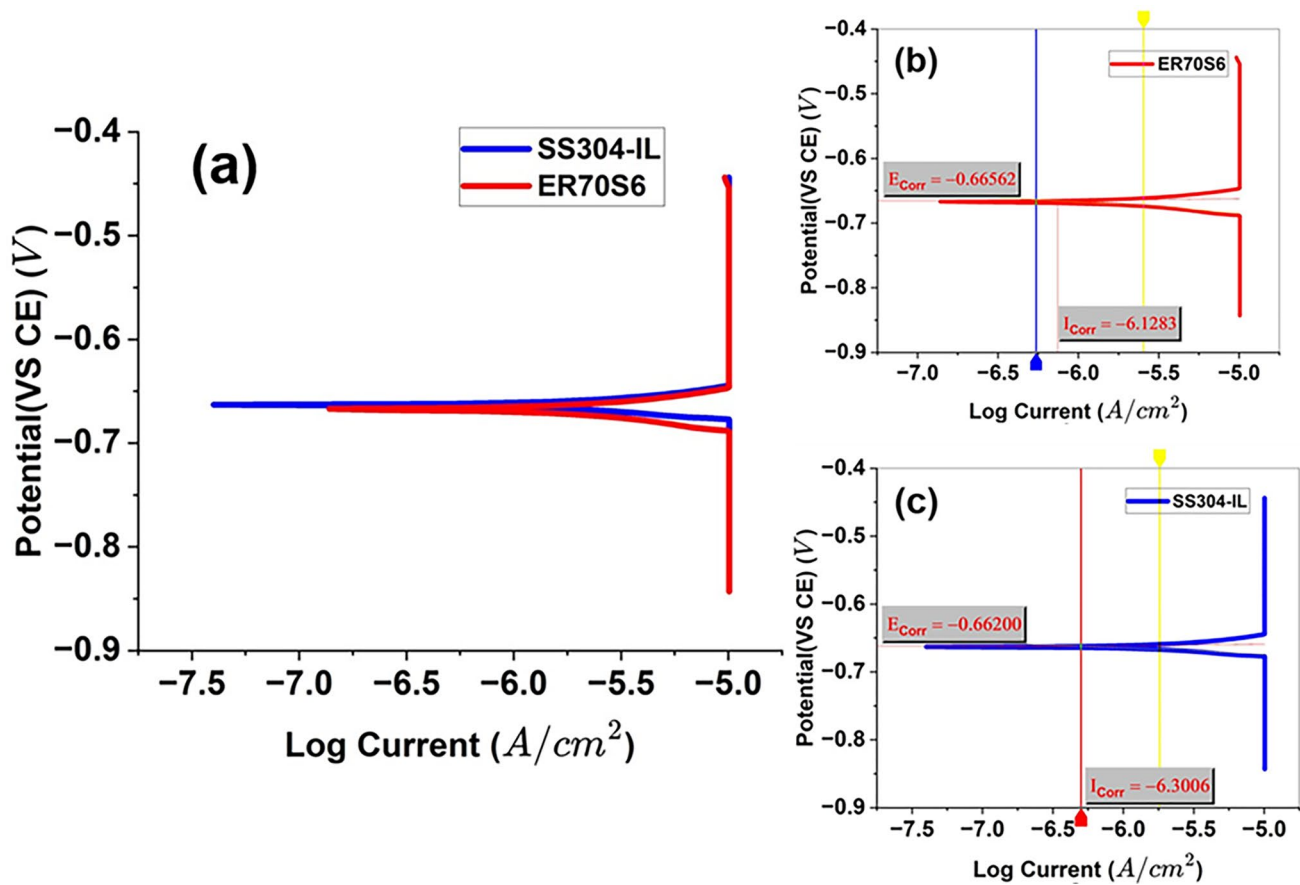


Fig. 11 a Potentiodynamic curves of the prepared samples b Tafel analysis result ER70S6 specimen c Tafel analysis result SS304-IL specimen

Conclusions

In this study, two ER70S6 samples were prepared. The first sample involved depositing copper-coated ER70S6 onto a mild steel substrate. For the second sample, a thin SS 304 foil (30 μm) was introduced as an interlayer between each printed layer. The microstructure, mechanical properties, and electrochemical corrosion performance of both samples were investigated and compared. The key findings of this study can be summarized as follows:

- OES and EDS analyses revealed the presence of chromium and nickel in the SS304-IL specimen, indicating successful fusion between the SS 304 interlayer and the ER70S6 mild steel base material. Thus, adding the interlayer could be treated as a convenient method of incorporating alloying elements which leads to improvement in overall properties including the corrosion resistance.
- XRD analysis showed that the specimen predominantly consists of an α -Iron (Body-Centered Cubic, BCC, Ferrite) phase, with peaks observed at 2θ values of approximately 44.67°, 65.02°, and 82.34°. Additionally, secondary phases such as Fe_2Si , Fe_3Si and Fe_3Ni were detected, notably in the SS304-IL specimens
- Microstructural analysis revealed that both specimens are primarily composed of ferrite with a small amount of lamellar pearlite. The ferrite content was slightly higher in the SS 304 interlayer specimen (81.74%) compared to the standard specimen (80.11%). The ferrite in the interlayer specimen was more uniform, and its grain size was slightly larger (9.21 μm) compared to the standard specimen (8.52 μm).
- Density measurements indicated that both specimens were dense, with the SS304-IL interlayer specimen having a higher density (7700 kg/m^3) compared to the standard specimen (7672 kg/m^3). The addition of the SS 304 interlayer also reduced porosity, from 1.84 % to 1.55 % area density.
- Tensile testing showed that the ultimate tensile strength of the specimens was generally higher in the horizontal direction compared to vertical and diagonal directions. The anisotropy of the SS304-IL specimen (3.54) was sig-

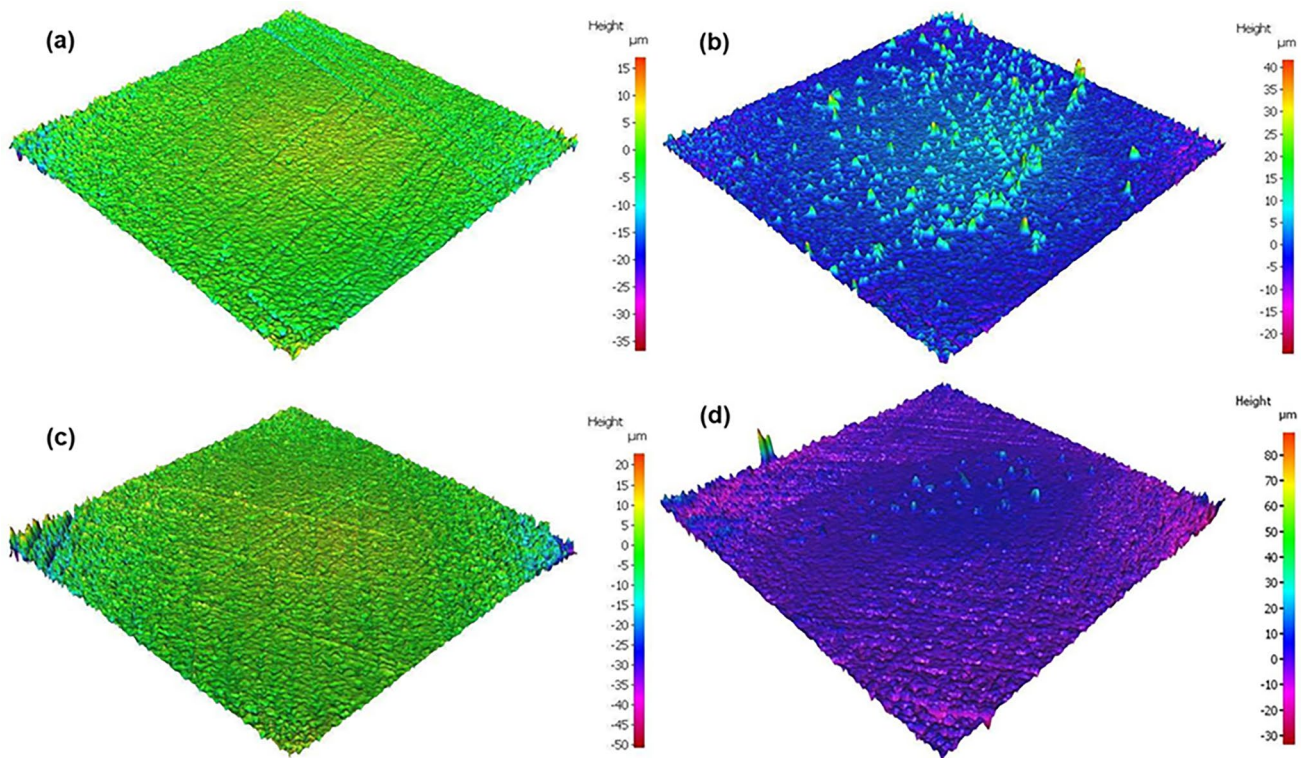


Fig. 12 Optical images of surfaces: **a** polished ER70S6, **b** ER70S6 after corrosion, **c** polished SS304-IL, **d** SS304-IL after corrosion

nificantly lower than that of the standard ER70S6 specimen (12.41), demonstrating that the interlayer reduced anisotropy in the WAAM specimen.

- Vickers hardness testing showed that the ER70S6 specimen exhibits a relatively stable hardness trend due to uniform heat dissipation, whereas the SS304-IL specimen shows greater fluctuations caused by localized heat accumulation and inhomogeneous mixing. These variations highlight the influence of the SS 304 interlayer on thermal effects and microstructural evolution in WAAM
- Electrochemical testing, performed in a 3.5 wt.% NaCl solution, revealed that the ER70S6 specimen had a lower E_{corr} but higher I_{corr} compared to the SS304-IL specimen, indicating a higher susceptibility to corrosion. The protective effect of the SS 304 interlayer was evident, resulting in a reduced corrosion rate for the interlayer specimen.

This study demonstrates the potential of incorporating an SS 304 interlayer in WAAM to enhance mechanical properties, reduce porosity, and improve corrosion resistance without modifying existing WAAM setups. The reduction in anisotropy and the observed improvements in microstructural uniformity highlight the efficacy of interlayer addition in mitigating typical WAAM-related defects. These findings

provide valuable insights for further optimizing WAAM-produced components for industrial applications where mechanical reliability and corrosion resistance are critical. Future research can explore different interlayer materials and thicknesses to further refine the technique and expand its applicability in advanced manufacturing processes

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Data availability All data that support the findings of this study are included within the article (and any supplementary files).

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